

Machine Learning Lab Assignment

Topic: K-Means Clustering (Unsupervised Learning)

Date: 27-01-2026

1. Apply K-means clustering for the following one-dimensional data points:

$X = \{2, 4, 10, 12\}$. Assume the number of clusters $K = 2$ and the initial centroids as

$c_1 = 2, c_2 = 10$. Perform the algorithm step by step using Euclidean distance and determine

1. Cluster assignment for each data point
 2. Updated centroids after the first iteration
 3. Final clusters after convergence
- 2.

Customer	Income	Score
c1	2	20
c2	3	25
c3	4	30
c4	5	35
c5	6	40
c6	8	65
c7	9	70
c8	10	75
c9	11	80
c10	12	85
c11	15	45
c12	16	50
c13	17	55
c14	18	60
c15	19	65

Apply the K-means clustering algorithm to cluster the above dataset into three clusters ($K = 3$). Assume the initial centroids as: $c_1 = (2, 20)$, $c_2 = (8, 65)$, $c_3 = (15, 45)$, perform the following steps

1. Compute Euclidean distances
2. Assign data points to clusters
3. Update centroids after the first iteration
4. Determine the final clusters after convergence